

CHAEYOON SHIN

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RESEARCH INTERESTS

- Leveraging diverse biomedical data (multi-omics, clinical registry, patient-generated health data) through computational & statistical modeling
- Quantifying heterogeneity in individual health trajectories and outcomes (e.g., causal machine learning, risk stratification, individualized intervention)
- Mental health & psychiatric phenotypes

EDUCATION

Sungkyunkwan University, Samsung Advanced Institute for Health Sciences & Technology (SAIHST) Mar 2026 –
M.S. in Digital Health
Advisor: Dr. Yoonjung Yoonie Joo

Sungkyunkwan University (SKKU) Mar 2022 – Feb 2026
B.E. in Integrative Biotechnology (*Magna Cum Laude*)

The University of Texas at Dallas Aug 2024 – Dec 2024
Fall Exchange Student Program

PUBLICATIONS

Preprints

- [1] **Chaeyoon Shin**, Junghoon Park, et al. Psychological resources confer greatest protection against late-life depression in genetically vulnerable adults. Preprint, June 2026. <https://doi.org/10.21203/rs.3.rs-9990161/v1>
- [2]

RESEARCH EXPERIENCE

Undergraduate Research Intern Jan 2025 – Feb 2026
Genomics & Neural Network Intelligence for Healthcare (GENNIE) Lab, SKKU

- Analyzed key determinants and their interaction with polygenic scores (GxE) on late-life depressive symptoms using the Wisconsin Longitudinal Study (WLS); applied H2O AutoML, moderation analysis, and generalized structured component analysis
- Examined heterogenous treatment effects of latency prolongation on neonatal outcomes following preterm premature rupture of membranes (PPROM) using Korean Neonatal Network (KNN); applied causal forest machine learning (GRF) and restricted cubic spline analysis

Undergraduate Research Program (URP) Dec 2023 – Aug 2024
Computational Cognitive Affective Neuroscience (COCOAN) Lab, SKKU

- Conducted fMRI data analysis and derived activation maps using MATLAB
- Participated in pilot studies and assisted fMRI experiments

Industry-Academia Collaboration Project Jan – Feb 2024
Lab of Molecular Development and Cell Signaling, SKKU

- Trained in cell culture and experimental techniques in ovarian cancer research
- Performed JC-1 staining, hanging drop assays, and tissue preparation protocols

CONFERENCE PRESENTATIONS

Poster presentation, Korean Society of Medical Informatics (KOSMI)

2026

(co-authored paper awarded Best Poster)

TEACHING & MENTORING

TA, <i>Introduction to Healthcare Data Analytics</i>	Spring 2026
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AWARDS & HONORS

<i>Scholarships</i>	
SKKU Graduate Scholarship (100% tuition support for Master's Program)	2026
KorUS Educational Exchange Initiative for Youth in STEM (\$9,000/semester) <i>Issued by Korea Institute for Advancement of Technology (KIAT)</i>	2024
SKKU Next Generation Scholars Program (100% tuition support for 3 rd ,4 th undergraduate years)	2024
SKKU Academic Excellence Scholarship (top 10%)	2023

EXTRACURRICULARS

Member of SKKAI (Sungkyunkwan AI Association), Medical AI Team	Sep 2025 – Jan 2026
• Explainable AI (XAI) research (sparse autoencoders) on pathological foundation models	
Member of National Undergraduate Symposium on Biology	Aug – Sep 2023
• Delivered a group presentation on cellular senescence and immunometabolism	

SKILLS

Languages: English (TOEFL 107/120, TOEIC 960/990), Korean (Native)
Programming: R, Python, Linux/Bash